LECTURE 12
Constructing discrete formulae & Numerical Stability

Reference: J. C. Tannehill, D. A. Anderson & R. H. Pletcher, Computational Fluid Mechanics and Heat Transfer, 2nd ed., Taylor & Francis, 1997. Chapter 3, Section 3.6.

Wladimir E. Banda-Barragán

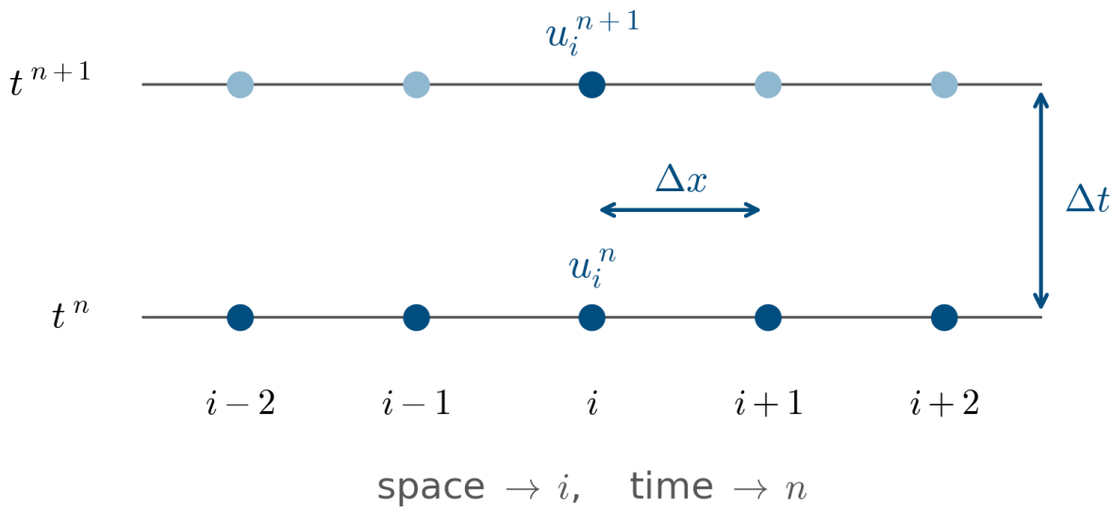
Notation convention: grid and indices

The space indices label the grid points, the marching index labels the time level:

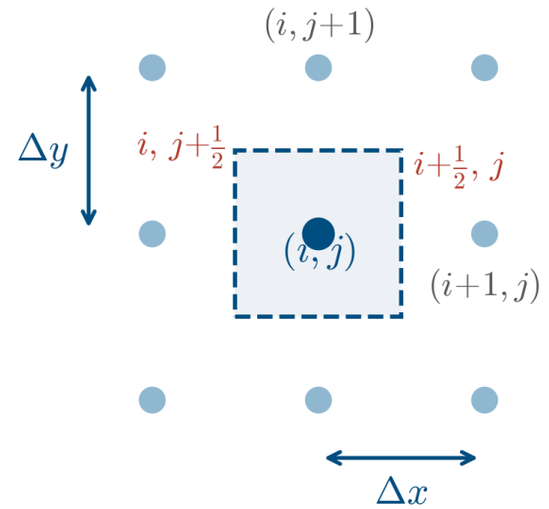
$$x \to i\,, \qquad y \to j\,, \qquad t \to n \qquad \Longrightarrow \qquad u(x,y,t) \longrightarrow u^n_{i,j}$$

which applies to 2D problems. In 1D, the notation is: u_i^n. In 3D, the notation is: $u^n_{i,j,k}$

(a) one space index and one time index



(b) two space indices



the dashed lines are the cell faces, at half-integer indices

In some analyses, it is useful to introduce a local coordinate: $\xi \equiv x - x_i$

$\xi = 0$ at the point itself, $\xi = \pm\Delta x$ at its neighbours

where ξ is x measured from the point we are working at.

Four ways to build an FDE from a PDE:

1. Taylor series:

We expand, combine, impose the order we want. This is the default way.

2. Polynomial fitting:

We fit a local polynomial and differentiate it. Used mostly at boundaries.

3. Integral method:

We integrate over a box in space and time, then approximate what is left.

4. Finite volume:

We apply the conservation law directly to a control volume.

The 4 ways often produce the same formula.

They do not produce the same way of thinking.

Route 1: Taylor series – systematic build

Goal:

We wish to approximate $\partial u/\partial x$ at i to $O[(\Delta x)^2]$ using only u at i, $i\,-\,1$ and $i\,-\,2$ — the situation at a boundary.

$$u_{i-2} = u_i + u_x(-2\Delta x) + u_{xx}\frac{(2\Delta x)^2}{2!} + u_{xxx}\frac{(-2\Delta x)^3}{3!} + \cdots$$

$$u_{i-1} = u_i + u_x(-\Delta x) + u_{xx}\frac{(\Delta x)^2}{2!} + u_{xxx}\frac{(-\Delta x)^3}{3!} + \cdots$$

$$u_x \equiv \left.\frac{\partial u}{\partial x}\right|_i , \qquad u_{xx} \equiv \left.\frac{\partial^2 u}{\partial x^2}\right|_i$$

We multiply the first by a, the second by b, and add them.
Then, we impose two conditions on the sum:

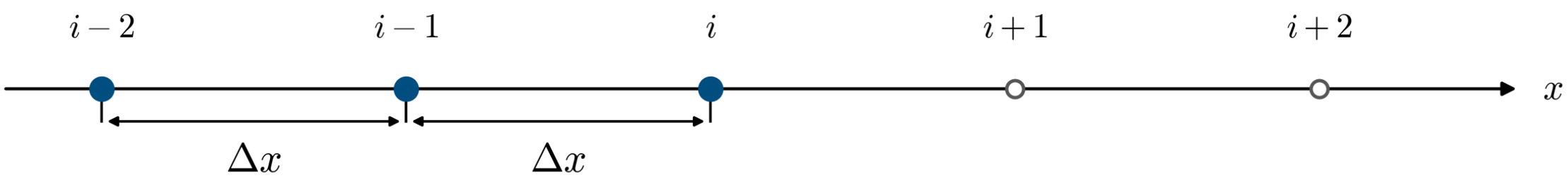
$$-2a - b = 1\,, \qquad 2a + \frac{b}{2} = 0 \qquad \Longrightarrow \qquad a = \tfrac{1}{2}\,, \quad b = -2$$

The result: second order, one side only

$$-2a - b = 1\,, \qquad 2a + \frac{b}{2} = 0 \qquad \Longrightarrow \qquad a = \tfrac{1}{2}\,, \quad b = -2$$

$$\boxed{\left.\frac{\partial u}{\partial x}\right|_i = \frac{u_{i-2} - 4u_{i-1} + 3u_i}{2\,\Delta x} + O\left[(\Delta x)^2\right]}$$

the only three points the formula may use



In a uniform mesh the formula reaches backwards only.

This is a one-sided, second order finite difference formula.
Useful at the (right) boundaries.

Run the same analysis backwards

Imagine someone hands you this difference formula from a paper or a code you inherited.

$$F \equiv \frac{u_{i-2} - 4u_{i-1} + 3u_i}{2\,\Delta x}$$

We can run the same analysis backwards to see what is actually computing and answer:

Does the error vanish under refinement? -> Consistency.
At what power of Δx? -> Order of accuracy

We substitute the two expansions of the previous slide and collect powers of Δx:

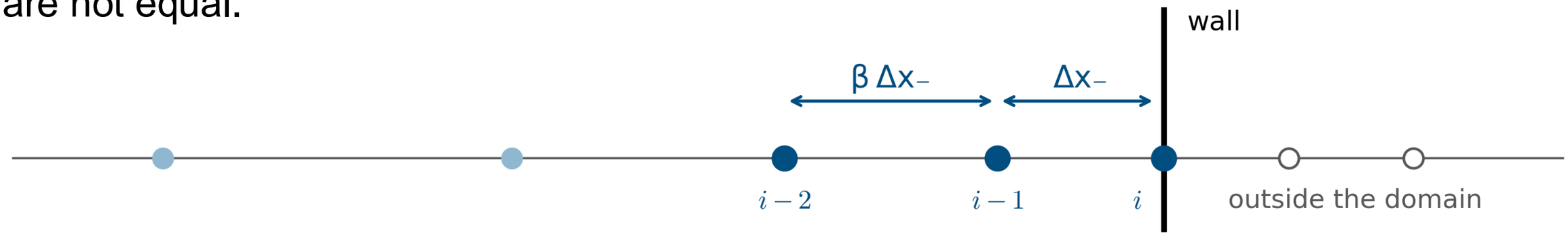
$$\begin{aligned} u_{i-2} - 4u_{i-1} + 3u_i \;&=\; \underbrace{(1-4+3)}_{0}\, u_i \;+\; \underbrace{(-2+4)}_{2}\, \Delta x\, u_x \\ &\quad + \underbrace{(2-2)}_{0} (\Delta x)^2 u_{xx} \;+\; \underbrace{\left(-\tfrac{8}{6}+\tfrac{4}{6}\right)}_{-2/3} (\Delta x)^3 u_{xxx} \;+\; \cdots \\ &=\; 2\,\Delta x\, u_x \;-\; \tfrac{2}{3}(\Delta x)^3 u_{xxx} \;+\; \cdots \end{aligned}$$

$$F = u_x - \tfrac{1}{3}(\Delta x)^2 u_{xxx} + O\left[(\Delta x)^3\right]$$

$$\boxed{\text{T.E.} = u_x - F = \tfrac{1}{3}(\Delta x)^2 u_{xxx} \;\longrightarrow\; 0 \;\text{ as }\; \Delta x \to 0} \qquad \Longrightarrow \text{ consistent, second order}$$

Non-uniform spacing at a wall

Imagine we have the same three points and the same procedure, but now the two gaps are not equal.



the mesh is clustered against the wall, so the gaps grow away from it (β > 1)

We must derive the formula for the mesh we actually have:

$$\Delta x_- \equiv x_i - x_{i-1}\,, \qquad \beta \equiv \frac{x_{i-1} - x_{i-2}}{x_i - x_{i-1}}$$

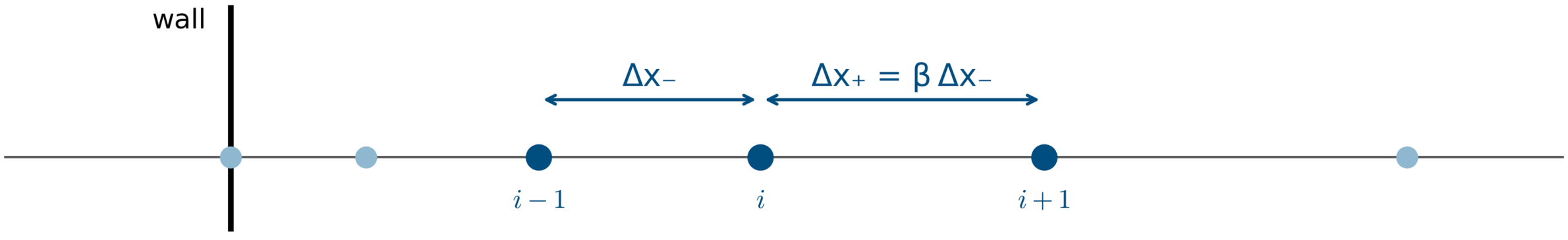
$$\left.\frac{\partial u}{\partial x}\right|_i = \frac{u_{i-2} - (1+\beta)^2\, u_{i-1} + \beta(2+\beta)\, u_i}{\beta(1+\beta)\,\Delta x_-} + O\left[(\Delta x)^2\right]$$

The weights change with β, but the order does not.

If $\beta = 1 \implies \dfrac{u_{i-2} - 4u_{i-1} + 3u_i}{2\,\Delta x}$ and we are back to the previous slide formula for uniform grids.

Non-uniform spacing at an interior point

The same stretched mesh, now seen from an interior point with a neighbour on each side.



the same mesh, now seen from an interior point: $\Delta x_+ \neq \Delta x_-$

The FDE is:

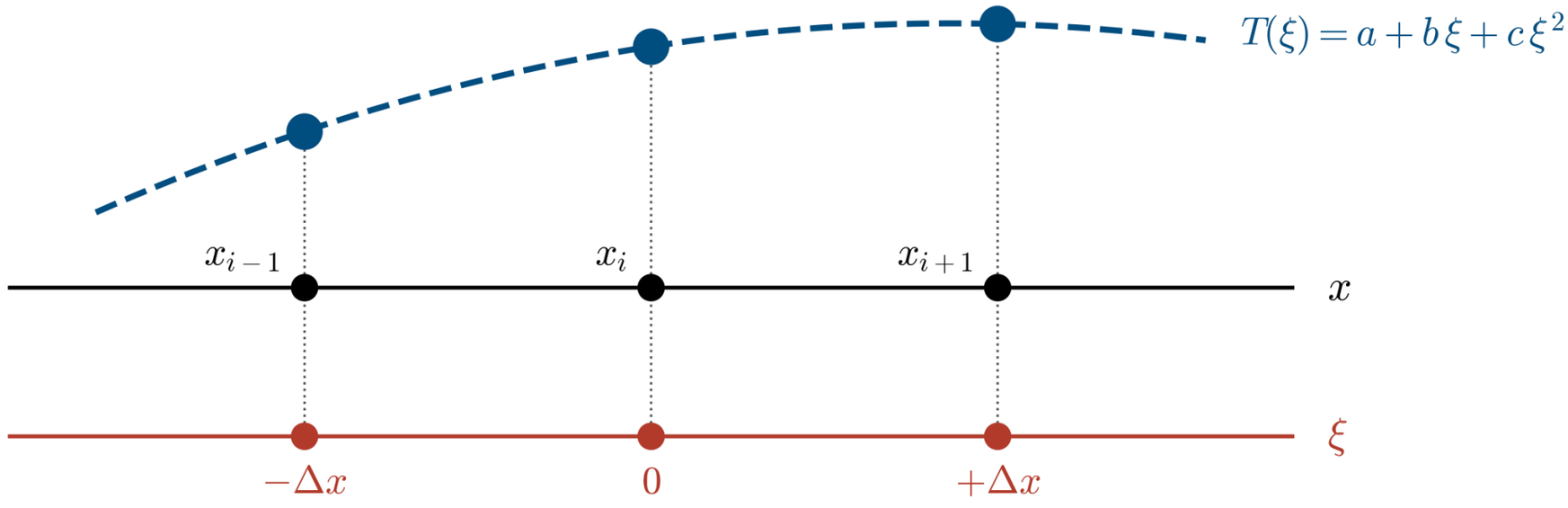
$$\left.\frac{\partial u}{\partial x}\right|_i = \frac{u_{i+1} + (\beta^2 - 1)\, u_i - \beta^2 u_{i-1}}{\beta(\beta+1)\,\Delta x_-} + O\left[(\Delta x)^2\right] \ , \qquad \beta \equiv \frac{\Delta x_+}{\Delta x_-}$$

If $\beta = 1 \implies \dfrac{u_{i+1} - u_{i-1}}{2\,\Delta x}$ and, we are back at the ordinary central difference.

Note:
We should keep β near 1: mesh generators warn above about 1.2.
Otherwise, we could revert back to first order accuracy.

Route 2: polynomial fitting

We assume the solution is locally a polynomial, fit it to the surrounding points, and differentiate the polynomial:



$\xi \equiv x - x_i$: the same axis, with its zero moved to the point itself

$$T(\xi) = a + b\,\xi + c\,\xi^2\,, \qquad \xi \equiv x - x_i \qquad \text{so } \xi = 0 \text{ at the point itself}$$

$$\Longrightarrow \qquad \left.\frac{\partial T}{\partial x}\right|_i = b\,, \qquad \left.\frac{\partial^2 T}{\partial x^2}\right|_i = 2c$$

With 3 equally spaced points this reproduces the central differences exactly. We use is at a wall:

$$\left.\frac{\partial T}{\partial x}\right|_i = \frac{1}{2\,\Delta x}\left(-3T_i + 4T_{i+1} - T_{i+2}\right) + O\left[(\Delta x)^2\right]$$

Route 3: the integral method

Let us consider the 1D heat equation for this route: $$\boxed{\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2}}$$

We integrate over one cell in x and one step in t.

The order of integration differs on the two sides, so each inner integral is exact:

$$\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \left(\int_{t^n}^{t^{n+1}} \frac{\partial u}{\partial t}\, dt \right) dx \;=\; \alpha \int_{t^n}^{t^{n+1}} \left(\int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \frac{\partial^2 u}{\partial x^2}\, dx \right) dt$$

The outer integrals have to be approximated, for which we use the **mean-value theorem**:

$$\text{if } f \text{ is continuous on } [y_1,\, y_1 + \Delta y] \quad \text{then some } \tilde{y} \text{ in that interval has } \int_{y_1}^{y_1+\Delta y} f(y)\, dy = f(\tilde{y})\, \Delta y$$

The theorem says $\tilde{y}$ exists, but it does not say where.

Notice that nothing has been approximated until this point (unlike in Taylor's route).

The choice of evaluation point

We apply the theorem on the right at some time t̃ inside the step, then write the 2 face gradients as differences:

$$\left[u_i^{n+1} - u_i^n\right] \Delta x = \alpha \left[\left.\frac{\partial u}{\partial x}\right|_{i+\frac{1}{2}} - \left.\frac{\partial u}{\partial x}\right|_{i-\frac{1}{2}} \right]_{\tilde{t}} \Delta t \qquad \tilde{t} \text{ is the one free choice}$$

$$\left.\frac{\partial u}{\partial x}\right|_{i+\frac{1}{2}} \simeq \frac{u_{i+1} - u_i}{\Delta x}, \qquad \left.\frac{\partial u}{\partial x}\right|_{i-\frac{1}{2}} \simeq \frac{u_i - u_{i-1}}{\Delta x}$$

Then we substitute and divide through by $\Delta x\, \Delta t$. The one free choice decides which scheme we get:

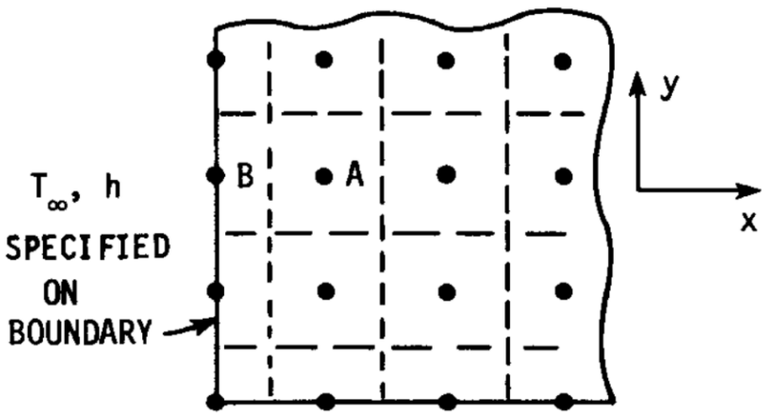
$$\tilde{t} = t^{n+1}: \quad \frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\alpha}{(\Delta x)^2} \left(u_{i+1}^{n+1} - 2u_i^{n+1} + u_{i-1}^{n+1}\right) \quad \text{implicit}$$

$$\tilde{t} = t^{n}: \quad \frac{u_i^{n+1} - u_i^n}{\Delta t} = \frac{\alpha}{(\Delta x)^2} \left(u_{i+1}^{n} - 2u_i^{n} + u_{i-1}^{n}\right) \quad \text{explicit}$$

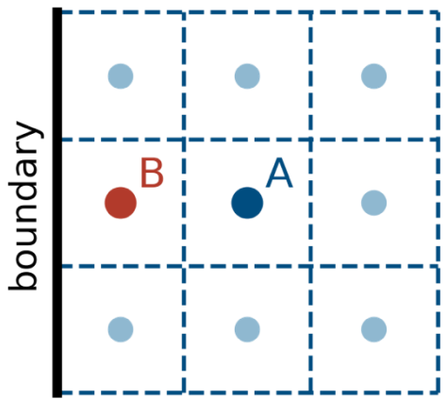
Nothing in the mathematics chooses t̃. The stability analysis does as we will see later.

Route 4: the finite-volume (FV) approach

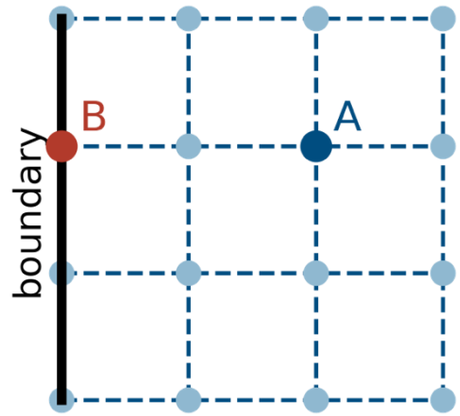
The other 3 routes accept the PDE and approximate its derivatives. This route goes back to the conservation statement. Let's see some definitions:



Cell-centred



Cell-vertex



The dots are the grid points, the dashed lines are the faces of the control volumes; A is an interior volume, B a boundary one.

Cell-centred: the volumes are drawn first, then a point at each centre.

Cell-vertex: the points are placed first, the faces halfway between.

Note:
On a stretched mesh the cell-vertex point is not at the centre of its own volume, so its value stops being the cell mean.

The integral conservation statement

Let us consider the 2D heat unsteady conduction equation for this route:

$$\boxed{\rho c \frac{\partial T}{\partial t} = \frac{\partial}{\partial x}\left(k \frac{\partial T}{\partial x}\right) + \frac{\partial}{\partial y}\left(k \frac{\partial T}{\partial y}\right)} \qquad \mathbf{q} = -k\nabla T$$

We integrate over the control volume V and apply the **divergence theorem**:

$$\iiint_V \left(\rho c \frac{\partial T}{\partial t} + \nabla \cdot \mathbf{q}\right) dV = 0 \qquad \xrightarrow{\text{divergence theorem}}$$

$$\underbrace{\iiint_V \rho c \frac{\partial T}{\partial t}\, dV}_{\text{energy stored}} + \underbrace{\oiint_S \mathbf{q} \cdot \mathbf{n}\, dS}_{\text{net efflux through the surface}} = 0$$

Therefore, the rate at which energy is stored in V equals the rate at which it is conducted in through the surface S (**conservation statement**).

Fluxes through the four faces

$$\mathbf{n}\, dS = \hat{\imath}\, dy - \hat{\jmath}\, dx \implies \iiint_V \rho c \frac{\partial T}{\partial t}\, dV + \oint_S (q_x\, dy - q_y\, dx) = 0$$

Valid for a volume of any shape.

For a rectangular cell, the contour integral becomes 4 face fluxes:

Storage: $$\rho c \frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t} \Delta x\, \Delta y$$

Four faces: $$- k\Delta y \left.\frac{\partial T}{\partial x}\right|_{i+\frac{1}{2},j} - k\Delta x \left.\frac{\partial T}{\partial y}\right|_{i,j+\frac{1}{2}} + k\Delta y \left.\frac{\partial T}{\partial x}\right|_{i-\frac{1}{2},j} + k\Delta x \left.\frac{\partial T}{\partial y}\right|_{i,j-\frac{1}{2}}$$

In adjacent cells, we have the same expression with opposite sign, so it cancels the sum:

$$\text{cell } (i,j): \quad -k\,\Delta y \left.\frac{\partial T}{\partial x}\right|_{i+\frac{1}{2},j} \qquad \text{cell } (i+1,j): \quad +k\,\Delta y \left.\frac{\partial T}{\partial x}\right|_{i+\frac{1}{2},j}$$

And the result is familiar > Explicit Scheme

Approximate the face gradients by central differences at level n:

$$\left.\frac{\partial T}{\partial x}\right|_{i+\frac{1}{2},\,j} \simeq \frac{T^n_{i+1,j} - T^n_{i,j}}{\Delta x}\,, \qquad \left.\frac{\partial T}{\partial y}\right|_{i,\,j+\frac{1}{2}} \simeq \frac{T^n_{i,j+1} - T^n_{i,j}}{\Delta y}$$

Substitute, divide through by $\rho c\ \Delta x\ \Delta y$, and rearrange:

$$\frac{T^{n+1}_{i,j} - T^n_{i,j}}{\Delta t} = \alpha \left[\frac{T^n_{i+1,j} - 2T^n_{i,j} + T^n_{i-1,j}}{(\Delta x)^2} + \frac{T^n_{i,j+1} - 2T^n_{i,j} + T^n_{i,j-1}}{(\Delta y)^2} \right]$$

Where: $\alpha = \dfrac{k}{\rho c}$

And we have found the explicit scheme for the 2D heat equation.

Notice we reached this without ever differencing a derivative. Instead, we used the FV approach.

What about the boundaries?

For the boundary volume B the conservation statement is unchanged.
Only one face is computed differently:

Interior face: $$q = -k \left.\frac{\partial T}{\partial x}\right|_{i-\frac{1}{2},j} \simeq -k \frac{T_{i,j} - T_{i-1,j}}{\Delta x}$$

Boundary face: $$q = h\left(T_\infty - T_{i,j}\right)$$

A wall heat flux, an adiabatic wall, a symmetry plane: all prescribed face fluxes.

The BC enters as a flux, in the same place and the same way as every interior flux.

The method-of-lines view: Discretise space to obtain ODEs in time, then choose a time integrator — 2 independent choices.

$$\frac{\partial \mathbf{U}}{\partial t} + \nabla \cdot \mathbf{F}(\mathbf{U}) = 0 \qquad \longrightarrow \qquad \frac{\mathbf{U}_i^{n+1} - \mathbf{U}_i^n}{\Delta t} + \frac{\mathbf{F}_{i+1/2}^n - \mathbf{F}_{i-1/2}^n}{\Delta x} = 0$$

Summary of the 4 routes

We have looked into how we can generate the same FDE in four different ways:

Route	Starts from	Gives the T.E.?	Natural for
Taylor series	the PDE, term by term	yes, automatically	anything on a structured mesh; the default
Polynomial fitting	assumed local form of the solution	no — compute it separately	boundaries, wall gradients, interpolation
Integral method	the PDE, integrated over a box	no — compute it separately	showing where explicit/implicit is decided
Finite volume	the *integral* conservation law	no — compute it separately	conservation, discontinuities, unstructured meshes

All 4 routes can give the same algebra.

Only the finite volume (FV) is built out of fluxes through faces, so only the finite volume way conserves by construction.

Stability

Stability concerns the error already present at a time level, and what the marching procedure does to it as the calculation proceeds:

$$\varepsilon_i^n \equiv N_i^n - D_i^n$$

D_i^n : exact solution of the difference equation

N_i^n : solution obtained in finite-precision arithmetic

$$\boxed{\text{stable} \iff |\varepsilon_i^n| \text{ remains bounded as } n \to \infty}$$

1. von Neumann analysis: places one Fourier mode in; sees the amplification factor out.

2. Domain of dependence: checks the stencil against the characteristic.

3. Taylor series: answers: what PDE does the scheme actually solve?

By **Lax's equivalence theorem**, a consistent scheme converges if and only if it is stable.

1. von Neumann analysis: the Fourier method

Expand the error in a Fourier series. The scheme is linear, so it is enough to follow one mode and ask what one step does to it:

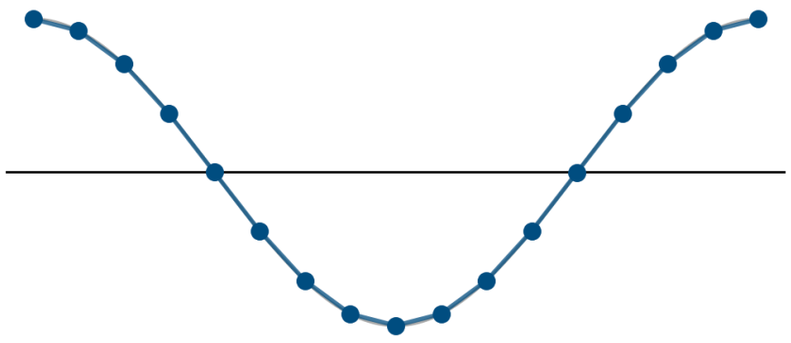
$$\varepsilon_i^n = G^n\, e^{\mathrm{I}\kappa x_i}\,, \qquad \beta \equiv \kappa\,\Delta x\,, \qquad G = \frac{\varepsilon_i^{n+1}}{\varepsilon_i^n}$$

$$\kappa_{\max} = \frac{\pi}{\Delta x} \quad \Longrightarrow \quad 0 \le \beta \le \pi$$

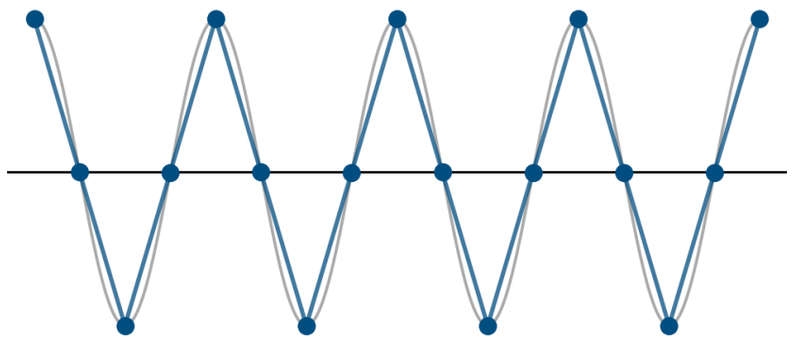
the sawtooth $+1, -1, +1, \ldots$ is the shortest wave the mesh carries

$$\boxed{|G| \le 1 \ \text{ for every } \beta \quad \Longleftrightarrow \quad \text{stable}}$$

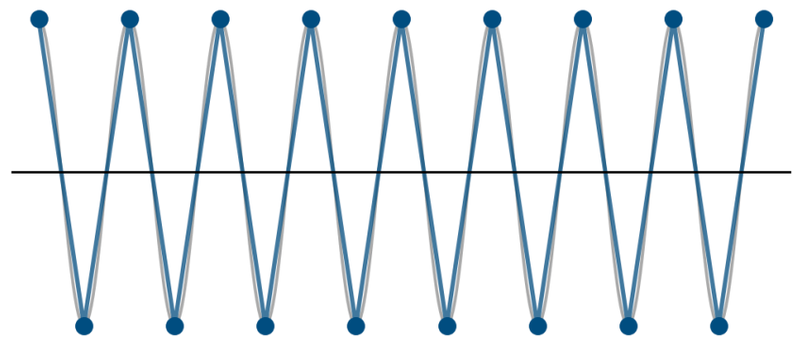
$\beta = \pi/8$ (smooth)



$\beta = \pi/2$



$\beta = \pi$ (sawtooth)



One mode sampled on the mesh. Smooth on the left; on the right β = π, the shortest wave the grid can carry.

von Neumann analysis on FTCS: heat equation

Let us consider the 1D heat equation for this route: $\dfrac{\partial T}{\partial t} = \alpha \dfrac{\partial^2 T}{\partial x^2}$

The FTCS scheme is linear, so the error obeys the same equation as T.

$$T_i^{n+1} = (1-2r)\,T_i^n + r\left(T_{i-1}^n + T_{i+1}^n\right), \qquad r = \frac{\alpha\,\Delta t}{(\Delta x)^2}$$

We substitute the mode: $\varepsilon_i^n = G^n\, e^{\mathrm{I}\kappa x_i}$

$$G^{n+1} e^{\mathrm{I}\kappa x_i} = G^n \left[(1-2r)\, e^{\mathrm{I}\kappa x_i} + r\left(e^{\mathrm{I}\kappa x_{i-1}} + e^{\mathrm{I}\kappa x_{i+1}}\right)\right]$$

$$x_{i\pm1} = x_i \pm \Delta x \quad \Longrightarrow \quad e^{\mathrm{I}\kappa x_{i\pm1}} = e^{\mathrm{I}\kappa x_i}\, e^{\pm \mathrm{I}\beta}$$

$$\text{divide by } G^n e^{\mathrm{I}\kappa x_i}: \qquad G = 1 - 2r + r\left(e^{\mathrm{I}\beta} + e^{-\mathrm{I}\beta}\right) = 1 - 2r + 2r\cos\beta$$

$$\cos\beta - 1 = -2\sin^2\tfrac{\beta}{2} \quad \Longrightarrow \quad G = 1 - 4r\sin^2\frac{\beta}{2}$$

$$|G| \le 1: \qquad -1 \;\le\; 1 - 4r\sin^2\tfrac{\beta}{2} \;\le\; 1 \qquad \text{worst case } \sin^2\tfrac{\beta}{2} = 1 \;\; (\beta = \pi)$$

Stability condition:

$$\boxed{r = \frac{\alpha\,\Delta t}{(\Delta x)^2} \le \frac{1}{2}} \quad \Longleftrightarrow \quad \Delta t \le \frac{(\Delta x)^2}{2\alpha}$$

von Neumann analysis on BTCS

Same 1D heat equation: $\dfrac{\partial T}{\partial t} = \alpha \dfrac{\partial^2 T}{\partial x^2}$

The spatial terms sit at level $n + 1$, so each of them carries a factor G:

$$-r\,T_{i-1}^{n+1} + (1+2r)\,T_i^{n+1} - r\,T_{i+1}^{n+1} = T_i^n$$

$$G\left[(1+2r) - r\left(e^{\mathrm{I}\beta} + e^{-\mathrm{I}\beta}\right)\right] = 1 \quad \Longrightarrow \quad G = \frac{1}{1 + 4r\sin^2\frac{\beta}{2}}$$

$$\text{the denominator is } \geq 1 \text{ for every } \beta \text{ and every } r > 0 \quad \Longrightarrow \quad |G| \leq 1 \text{ always}$$

Unconditionally stable: no condition on r at all.

If you want to see the full construction to the FDE formulae, see:

FTCS: https://github.com/wbandabarragan/computational-physics-2/blob/main/unit-4/405-PDE-heat-equation-explicit.ipynb
BTCS: https://github.com/wbandabarragan/computational-physics-2/blob/main/unit-4/406-PDE-heat-equation-implicit.ipynb
Crank-Nicolson: https://github.com/wbandabarragan/computational-physics-2/blob/main/unit-4/407-PDE-Crank-Nicolson.ipynb

von Neumann analysis on Crank–Nicolson

Same 1D heat equation: $\dfrac{\partial T}{\partial t} = \alpha \dfrac{\partial^2 T}{\partial x^2}$

Half the spatial operator sits at level n and half at $n + 1$, so half of it carries a factor G:

$$-r\,T_{i-1}^{n+1} + 2(1+r)\,T_i^{n+1} - r\,T_{i+1}^{n+1} = r\,T_{i-1}^{n} + 2(1-r)\,T_i^{n} + r\,T_{i+1}^{n}$$

$$G\left[2 + 4r\sin^2\tfrac{\beta}{2}\right] = 2 - 4r\sin^2\tfrac{\beta}{2} \quad \Longrightarrow \quad \boxed{G = \frac{1 - 2r\sin^2\frac{\beta}{2}}{1 + 2r\sin^2\frac{\beta}{2}}}$$

$$\text{numerator} \le \text{denominator in modulus} \quad \Longrightarrow \quad |G| \le 1 \text{ for every } r > 0$$

Comparing this result with the implicit (BTCS) scheme:

$$r \to \infty : \quad G_{\text{implicit}} \to 0\,,$$

$$G_{\text{C-N}} \to -1 \quad \text{(modulus 1: it flips sign every step and never decays)}$$

Please note that unconditional stability does NOT translate to accuracy:
the truncation error is unchanged, so a large step is stable and wrong.

Other methods:

Richardson's method: central in time as well as in space.
Second order in both, so by the measures of Lecture 11 it should be the best of the four:

$$\frac{u_i^{n+1} - u_i^{n-1}}{2\,\Delta t} = \frac{\alpha}{(\Delta x)^2}\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right), \qquad \text{T.E.} = O\left[(\Delta t)^2, (\Delta x)^2\right]$$

However, it is accurate but **unconditionally unstable**:

$$\frac{G^2 - 1}{2\,\Delta t} = \frac{\alpha}{(\Delta x)^2} G\left(e^{\mathrm{I}\beta} - 2 + e^{-\mathrm{I}\beta}\right) = -\frac{4\alpha}{(\Delta x)^2} G \sin^2\frac{\beta}{2}$$

$$G^2 + 2bG - 1 = 0\,, \qquad b \equiv 4r\sin^2\tfrac{\beta}{2} \geq 0 \quad \Longrightarrow \quad G_\pm = -b \pm \sqrt{b^2+1}$$

$$G_+ G_- = -1 \quad \Longrightarrow \quad \boxed{|G_-| = b + \sqrt{b^2+1} \;>\; 1 \;\; \text{for every } r > 0}$$

DuFort–Frankel method: solved this by replacing the centre value by the average of levels $n - 1$ and $n + 1$, and is stable for every r. However, it is conditionally consistent.

Truncation error and stability are independent properties.
Both have to be checked always!

Summary: stability of heat-equation schemes

Only the time levels on the right-hand side change:

Scheme	Difference equation	Truncation error, stability
FTCS	$\dfrac{u_i^{n+1}-u_i^n}{\Delta t}=\dfrac{\alpha}{(\Delta x)^2}\left(u_{i+1}^n-2u_i^n+u_{i-1}^n\right)$	$O[\Delta t,(\Delta x)^2]$ explicit, $r\le\frac{1}{2}$
BTCS	$\dfrac{u_i^{n+1}-u_i^n}{\Delta t}=\dfrac{\alpha}{(\Delta x)^2}\left(u_{i+1}^{n+1}-2u_i^{n+1}+u_{i-1}^{n+1}\right)$	$O[\Delta t,(\Delta x)^2]$ implicit, every r
Crank–Nicolson	$\dfrac{u_i^{n+1}-u_i^n}{\Delta t}=\dfrac{\alpha}{2(\Delta x)^2}\left[\left(u_{i+1}^n-2u_i^n+u_{i-1}^n\right)+\left(u_{i+1}^{n+1}-2u_i^{n+1}+u_{i-1}^{n+1}\right)\right]$	$O[(\Delta t)^2,(\Delta x)^2]$ implicit, every r
Richardson (CTCS)	$\dfrac{u_i^{n+1}-u_i^{n-1}}{2\,\Delta t}=\dfrac{\alpha}{(\Delta x)^2}\left(u_{i+1}^n-2u_i^n+u_{i-1}^n\right)$	$O[(\Delta t)^2,(\Delta x)^2]$ explicit, **never**
DuFort–Frankel	$\dfrac{u_i^{n+1}-u_i^{n-1}}{2\,\Delta t}=\dfrac{\alpha}{(\Delta x)^2}\left(u_{i+1}^n-u_i^{n+1}-u_i^{n-1}+u_{i-1}^n\right)$	$O[(\Delta t)^2,(\Delta x)^2]$, plus $(\Delta t/\Delta x)^2$ explicit, every r^*

* consistent only if $\Delta t/\Delta x\to 0$

Richardson has the best truncation error here and is never stable
DuFort–Frankel is always stable and only conditionally consistent.

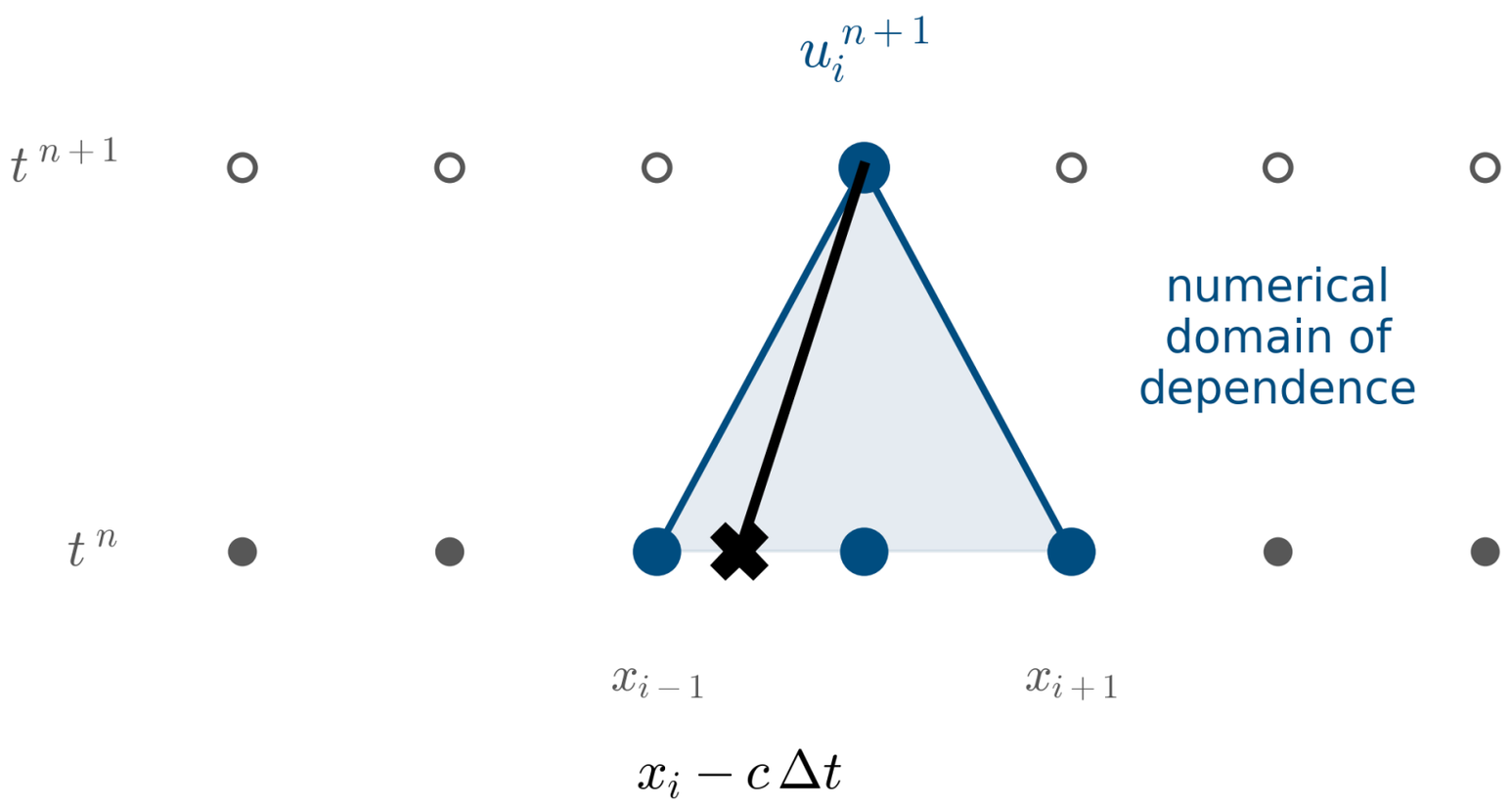
2. Domain of dependence: wave equation

The second stability test is geometric. It needs a finite speed of propagation, so we will use the **one-way wave equation (hyperbolic problem)**:

$$\frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = 0$$

This PDE just translates the profile, so a new value comes from one point:

$$u(x,t) = u_0(x - ct) \qquad \Longrightarrow \qquad u\left(x_i,\, t^{n+1}\right) = u(x_i - c\,\Delta t,\;\; t^{n})$$



Here the foot $x_i - c\Delta t$ falls between $i-1$ and i, inside the three points the scheme reads.

The stencil must contain the characteristic

The exact answer comes from one point: the foot at $x_\mathrm{i} - c\Delta t$.

The scheme reads only the three points $i - 1$, i, and $i + 1$.

Therefore, a one-step explicit scheme can only be right if the foot falls inside the interval it actually reads:

$$x_i - \Delta x \;\leq\; x_i - c\,\Delta t \;\leq\; x_i + \Delta x \qquad \Longleftrightarrow \qquad |c\,\Delta t| \leq \Delta x$$

This is what is known as the **CFL condition,** where the CFL number is defined as:

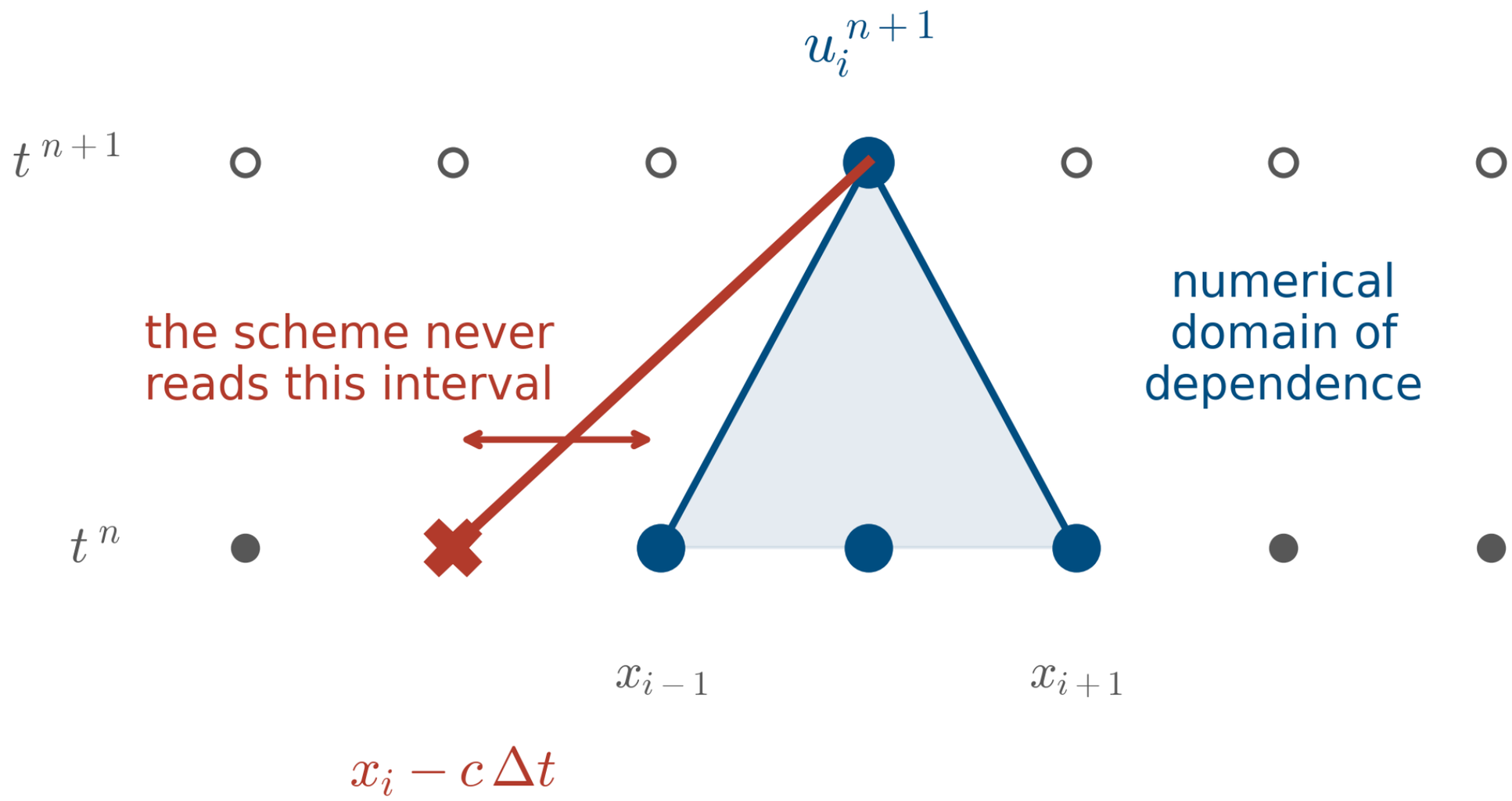
$$\boxed{\mathrm{CFL} \equiv \left|\frac{c\,\Delta t}{\Delta x}\right| \leq 1} \qquad \Longleftrightarrow \qquad \Delta t \leq \frac{\Delta x}{|c|}$$

The CFL number is named after Courant, Friedrichs and Lewy, 1928.

They found this condition for hyperbolic advection ~20 years before there was a computer to run it on.

What happens if the CFL condition is broken?

If $CFL > 1$, the foot of the characteristic falls outside the stencil, so the value that determines the answer is never read:



No arithmetic can recover information that the scheme did not read.
CFL is a necessary condition; it is not automatically sufficient.

The Courant (CFL) number:

The condition involves a dimensionless group, and it is best read as a ratio of 2 lengths:

$$\mathrm{CFL} \equiv \frac{c\,\Delta t}{\Delta x} = \frac{\overbrace{c\,\Delta t}^{\text{distance travelled in one step}}}{\underbrace{\Delta x}_{\text{width of one cell}}} = \text{cells crossed per step}$$

$CFL < 1$: the wave crosses less than one cell per step.

$CFL = 1$: exactly one cell; for the Lax method this is the exact answer.

$CFL > 1$: the scheme cannot see where the information came from.

In real codes designed for CFD, the CFL number is the number we set in the input file.

The time step is retrieved from the CFL number:

$$\Delta t = \mathrm{CFL}\,\frac{\Delta x}{\max|c|}$$

3. The Taylor-series test: FTCS

We use again the one-way wave equation:

$$\boxed{\frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = 0}$$

Then, we use the FTCS scheme to expand every term of the difference equation about (i, n) and see which PDE is left standing:

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} + c\,\frac{u_{i+1}^n - u_{i-1}^n}{2\,\Delta x} = 0$$

$$\Longrightarrow \quad \frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} + \frac{\Delta t}{2}\,\frac{\partial^2 u}{\partial t^2} + \frac{c\,(\Delta x)^2}{6}\,\frac{\partial^3 u}{\partial x^3} + \cdots = 0$$

The leading error term is a time derivative, so we use the two-way wave equation itself to turn it into a space derivative:

$$\frac{\partial^2 u}{\partial t^2} = c^2\,\frac{\partial^2 u}{\partial x^2} \quad \Longrightarrow \quad \frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = \alpha_{\text{eff}}\,\frac{\partial^2 u}{\partial x^2} + \cdots\,, \qquad \boxed{\alpha_{\text{eff}} = -\,\frac{c^2 \Delta t}{2} < 0}$$

A diffusion with a negative coefficient is an amplifier.
FTCS on the wave equation is unstable for every Δt.

The Lax method: averaging leads to stability

Lax replaces u at i by the average of its two neighbours:

$$u_i^{n+1} = \underbrace{\frac{u_{i+1}^n + u_{i-1}^n}{2}}_{\text{the average, in place of } u_i^n} - \frac{\mathrm{CFL}}{2}\left(u_{i+1}^n - u_{i-1}^n\right)$$

That substitution is an extra term in disguise: **positive diffusion**

Repeating the same Taylor series expansion now gives:

$$\frac{\partial u}{\partial t} + c\,\frac{\partial u}{\partial x} = \alpha_{\text{eff}}\,\frac{\partial^2 u}{\partial x^2} + \cdots, \qquad \alpha_{\text{eff}} = \frac{(\Delta x)^2}{2\,\Delta t}\left(1 - \mathrm{CFL}^2\right)$$

$$\alpha_{\text{eff}} \geq 0 \quad \Longleftrightarrow \quad \boxed{|\mathrm{CFL}| \leq 1}$$

The averaging injects positive diffusion, enough to acquire stability, provided that $|CFL| \leq 1$ — the same condition found earlier, by an independent route.

Dissipation and dispersion

Note also that for a hyperbolic problem G comes out complex:

$$G = |G|\, e^{\mathrm{I}\phi}$$ which has modulus and phase.

We can compare them with the exact factor. One exact step of the wave equation multiplies the mode by:

$$G_{\text{exact}} = e^{-\mathrm{I}\kappa c\,\Delta t} = e^{-\mathrm{I}\,\mathrm{CFL}\,\beta} \qquad \Longrightarrow \qquad |G_{\text{exact}}| = 1$$

$$\phi_{\text{exact}} = -\,\mathrm{CFL}\,\beta$$

Dissipation:

If $|G| < 1$, the amplitude is lost, and the profile smears.

Dispersion:

If $\varphi \neq -CFL\,\beta$, the mode travels at the wrong speed, and the profile wobbles.

Dispersion needs no instability:
every mode can stay bounded and still arrive out of step with the others.

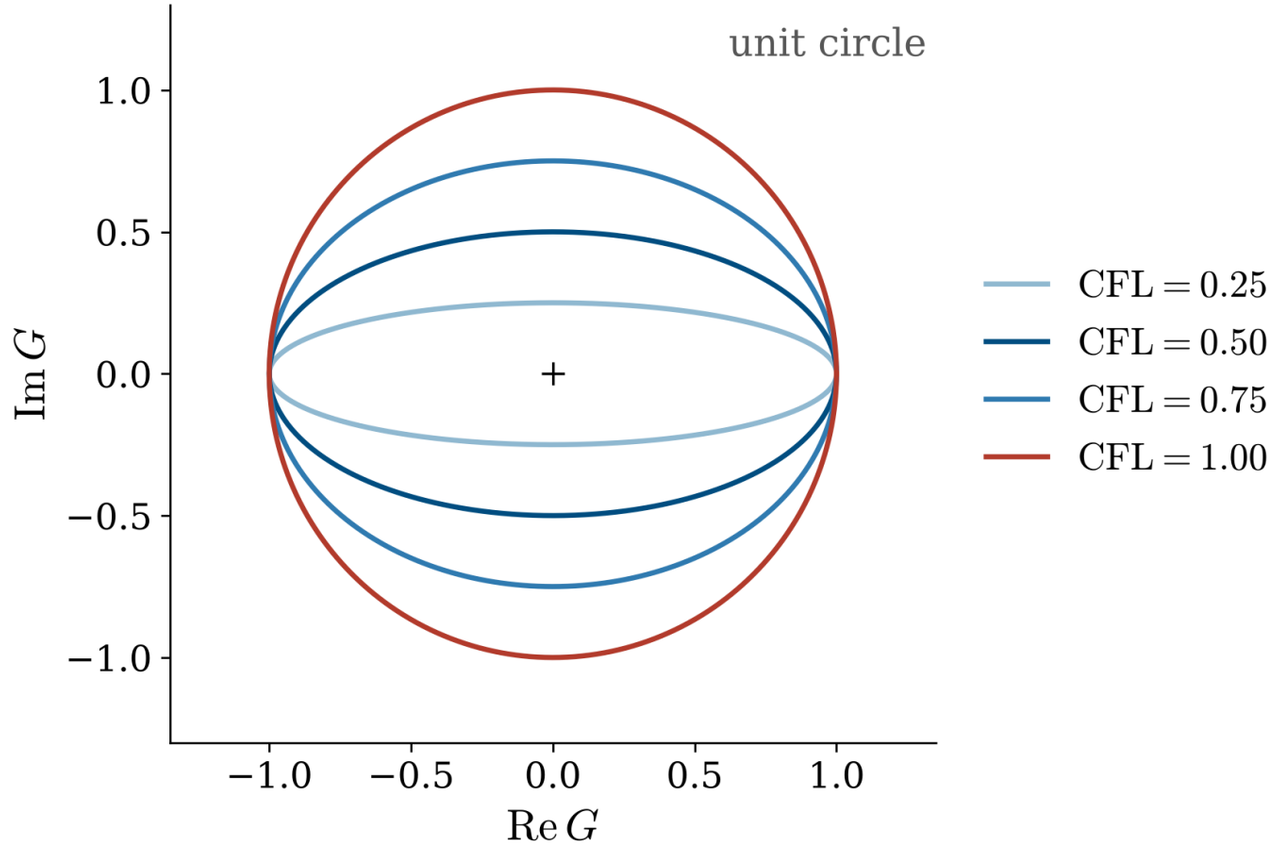
The Lax factor in the complex plane

We place one mode into the Lax scheme and divide through by the spatial factor:

$$G = \frac{e^{\mathrm{I}\beta} + e^{-\mathrm{I}\beta}}{2} - \frac{\mathrm{CFL}}{2}\left(e^{\mathrm{I}\beta} - e^{-\mathrm{I}\beta}\right) = \boxed{\cos\beta - \mathrm{I}\,\mathrm{CFL}\,\sin\beta}$$

$$|G|^2 = \cos^2\beta + \mathrm{CFL}^2\sin^2\beta = 1 - \left(1 - \mathrm{CFL}^2\right)\sin^2\beta$$

We obtain the equation of an ellipse in the complex plane:



The locus over 0 ≤ β ≤ 2π: an ellipse with semi-axes 1 and CFL.

Additional cases: 2D and systems of equations

2D cases: we have one mode and one phase angle per direction; nothing else changes:

$$\varepsilon_{i,j}^{n} = G^{n}\, e^{\mathrm{I}\kappa_x x_i}\, e^{\mathrm{I}\kappa_y y_j} \quad \Longrightarrow \quad G = 1 - 4r_x \sin^2\frac{\beta_x}{2} - 4r_y \sin^2\frac{\beta_y}{2}$$

$$\boxed{r_x + r_y \leq \tfrac{1}{2}} \qquad \Delta x = \Delta y \quad \Longrightarrow \quad r \leq \tfrac{1}{4} \quad \text{(half the 1-D step)}$$

Systems of equations: the error becomes a vector, so G is a matrix that mixes and scales the components:

$$\boldsymbol{\varepsilon}^{n+1} = [G]\,\boldsymbol{\varepsilon}^{n} \quad \Longrightarrow \quad \boldsymbol{\varepsilon}^{n} = [G]^{n}\,\boldsymbol{\varepsilon}^{0}$$

$$\boxed{|\lambda_\ell| \leq 1 + O(\Delta t) \ \text{ for every eigenvalue } \lambda_\ell \text{ of } [G]}$$

Non-linear cases: for a nonlinear system, the Jacobian is frozen over one step, and the same analysis is performed. It represents a local result, used as a guide.

Limitations of the von Neumann analysis

Boundaries:
The Fourier expansion assumes periodicity, so the result describes the interior of the mesh only.

Variable coefficients:
The spatial factor cancels only because the coefficients are constant. The limit is local.

Matrix method:
Write one step as a matrix acting on the whole solution vector, boundary rows included:

$$\mathbf{u}^{n+1} = [X]\,\mathbf{u}^{n} \qquad \text{(the boundary rows are part of } [X]\text{)}$$

$$\text{stable} \iff |\lambda_\ell| \le 1 \quad \text{for every eigenvalue of } [X]$$

Thus, the limit this analysis gives is an upper bound on the true one, and it is used with a margin.

Why is the time step set below the limit?

$$\Delta t = \mathrm{CFL}\,\Delta t_{\max}\,, \qquad \mathrm{CFL} \approx 0.8\text{–}0.9$$

The limit is local:
It is evaluated from the current state at sampled points. The worst cell may not be one of them.

The state changes:
In a nonlinear problem the wave speed changes during the very step you are taking.

Two effects are missing:
Boundaries and non-linearity, as the previous slide showed. Both tighten the limit.

The CFL number we set is not a fudge factor:
it covers the three assumptions the analysis had to make to give a closed-form answer.

The schemes and their stability limits

Everything proved in the second half, in one place:

Scheme	Amplification factor	Stable if
FTCS, heat	$1 - 4r\sin^2\frac{\beta}{2}$	$r \le \frac{1}{2}$
BTCS, heat	$\dfrac{1}{1 + 4r\sin^2\frac{\beta}{2}}$	every r
Crank–Nicolson	$\dfrac{1 - 2r\sin^2\frac{\beta}{2}}{1 + 2r\sin^2\frac{\beta}{2}}$	every r; $G \to -1$ as $r \to \infty$
Richardson	roots of $G^2 + 2bG - 1 = 0$	never
DuFort–Frankel	roots of $(1+2r)G^2 - 4rG\cos\beta - (1-2r) = 0$	every r (but consistent only if $\Delta t/\Delta x \to 0$)
FTCS, wave	$1 - \mathrm{I}\,\mathrm{CFL}\,\sin\beta$	never
Lax	$\cos\beta - \mathrm{I}\,\mathrm{CFL}\,\sin\beta$	$\lvert\mathrm{CFL}\rvert \le 1$

Notice the important factors:
$\boldsymbol{r = \alpha\Delta t/(\Delta x)^2}$ **for the parabolic problems**
$\boldsymbol{CFL = c\Delta t/\Delta x}$ **for the hyperbolic ones.**

Summary

Four routes to an FDE:
Taylor series, polynomial fitting, the integral method, finite volume.

Conservation? Only the finite-volume (FV) route is built from face fluxes, so only it conserves by construction.

Stability:
The error must not grow. Independent of accuracy, and the harder part of Lax's theorem.

G: the factor multiplying one Fourier mode per step.
Stable $\Leftrightarrow |G| \leq 1$ for every β in $[0, \pi]$.

Parabolic: G real; $r \leq \frac{1}{2}$ explicit, unconditional implicit; $\Delta t \sim (\Delta x)^2$.

Hyperbolic: G complex; $CFL \leq 1$; $\Delta t \sim \Delta x$. The modulus describes dissipation, the argument describes dispersion.

Our analysis leaves out the boundaries and the nonlinearities that may arise, so in practice we should run simulations at CFL numbers < 1.